

LAB 5

HTML5 Semantic Tags, Form Validation & Multimedia

• Definitions • Attributes • Examples • Output • Viva

1. HTML5 Semantic Tags

<header>

What is it? Introductory/top content.

Example:

```
<header>
  <h1>My Website</h1>
</header>
```

Output: A heading appears at the top.

💡 **header → introduction**

<nav>

What is it? Important navigation links.

Example:

```
<nav>
  <a href="#home">Home</a>
  <a href="#about">About</a>
</nav>
```

Output: Navigation links appear.

💡 **nav → navigation**

<main>

What is it? Main unique content of the page.

Example:

```
<main>
  <h2>Welcome</h2>
  <p>Main content.</p>
</main>
```

Output: Main content appears.

💡 **main → main content**

<article>

What is it? Self-contained content such as a blog/news post.

Example:

```
<article>
  <h2>HTML5</h2>
```

```
<p>Semantic HTML is useful.</p>
</article>
```

Output: An independent article appears.

 **article → independent content**

<section>

What is it? Groups related content under one topic.

Example:

```
<section>
  <h2>Services</h2>
  <p>Web Design</p>
</section>
```

Output: A related section appears.

 **section → related content**

<footer>

What is it? Footer information such as copyright or links.

Example:

```
<footer>
  <p>© 2026 Yug Sardhara</p>
</footer>
```

Output: Footer information appears.

 **footer → footer information**

Note:- &Copy; Used For Copyright Symbol ©

Complete Semantic Layout

```
<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#about">About</a>
</nav>

<main>
  <section>
    <h2>About Us</h2>
    <p>Welcome to our website.</p>
  </section>

  <article>
    <h2>Latest Article</h2>
    <p>Article content.</p>
  </article>
</main>
```

```
<footer>
  <p>© 2026 Yug Sardhara</p>
</footer>
```

💡 Global attributes such as **id**, **class**, **style** and **title** can also be used with these elements.

2. HTML5 Contact Form Elements

HTML5 provides special input types so the browser knows what kind of information the user should enter.

<input type="email">

What is it? Email address input.

Example:

```
<label for="email">Email:</label>
<input type="email" id="email" name="email" required>
```

Attributes:

Attribute	Simple meaning
type	Email input.
id	Unique identifier; can connect the label.
name	Name used when form data is submitted.
required	Makes it compulsory.
placeholder	Temporary hint.

Output: Email input appears.

<input type="tel">

What is it? Telephone/mobile input. It does not automatically require exactly 10 digits.

Example:

```
<input type="tel" id="phone" pattern="[0-9]{10}" placeholder="10-digit number">
```

Attributes:

Attribute	Simple meaning
type	Telephone input.
pattern	Custom phone format.
placeholder	Temporary hint.
required	Makes it compulsory.

Output: Phone input appears; pattern can require 10 digits.

<input type="date">

What is it? Date input; many browsers provide a date picker.

Example:

```
<input type="date" id="dob" min="2000-01-01" max="2026-12-31">
```

Attributes:

Attribute	Simple meaning
type	Date control.
min	Earliest allowed date.
max	Latest allowed date.
required	Makes it compulsory.

Output: Date field/date picker appears.

<textarea>

What is it? Multi-line text field for messages and feedback.

Example:

```
<textarea id="message" rows="5" cols="30" minlength="10" maxlength="200" required></textarea>
```

Attributes:

Attribute	Simple meaning
rows	Visible lines.
cols	Visible width.
minlength	Minimum characters.
maxlength	Maximum characters.
placeholder	Temporary hint.
required	Requires text.

Output: Multi-line text box appears.

3. required Attribute

required tells the browser that a value must be entered before successful form submission.

```
<form>
  <input type="text" name="name" required>
  <input type="email" name="email" required>
  <input type="tel" name="phone" required>
  <button type="submit">Submit</button>
</form>
```

If a required field is empty, the browser normally shows a validation message and prevents successful submission.

💡 **required = compulsory**

4. Registration Form Validation

Validation means checking whether entered data follows the rules we set.

type="number"

Creates a numeric input such as age, marks or quantity.

```
<input type="number" id="age" min="18" max="60" required>
```

Attribute	Simple meaning
min	Smallest allowed value.
max	Largest allowed value.
step	Allowed increment.
required	Compulsory.

Output: numeric input; values outside the range fail browser validation.

💡 **number → numbers | min/max → range**

pattern

pattern defines a regular-expression rule. The entered value must match that rule during form validation. It is useful for custom formats such as passwords.

```
<input type="password"
  pattern="(?=.*[A-Z])(?=.*[0-9]).{8,}"
  title="8+ chars, uppercase and number"
  required>
```

Attribute	Simple meaning
pattern	Regular expression the value must match.
title	Can explain the required format.

Output: password is checked against the pattern when submitted.

💡 **pattern = custom format/rule**

Email + Number + Pattern Together

```
<form>
  <input type="email" required>
  <input type="number" min="18" max="60" required>
  <input type="password"
    pattern="(?=.*[A-Z])(?=.*[0-9]).{8,}"
    title="8+ chars, uppercase and number" required>
  <button type="submit">Register</button>
</form>
```

5. Feedback Form: minlength, maxlength, min and max

minlength and maxlength control character count. min and max control a value range.

Attribute	Simple meaning
minlength	Minimum characters.
maxlength	Maximum characters.
min	Minimum value.
max	Maximum value.

```
<textarea minlength="10" maxlength="200"
  placeholder="Write feedback..." required></textarea>
```

```
<input type="number" min="1" max="5" required>
```

Output: feedback must contain 10–200 characters; the rating must be between 1 and 5.

💡 **pattern = FORMAT | minlength/maxlength = CHARACTERS | min/max = VALUE RANGE**

6. <audio> Tag

The <audio> element embeds sound in a webpage.

```
<audio controls autoplay loop>
  <source src="song.mp3" type="audio/mpeg">
  Your browser does not support audio.
</audio>
```

Attribute	Simple meaning
controls	Shows play, pause, volume and other controls.
autoplay	Requests automatic playback; browsers may block audible autoplay.
loop	Repeats after finishing.
muted	Starts without sound.
preload	Hints how much audio to load.
src	Audio URL; a source element can also provide it.

Output: an audio player appears. If autoplay is allowed, it may start automatically and loop.

💡 **controls = controls | autoplay = automatic start | loop = repeat**

7. <source> Tag

source provides the actual media file and its format. Multiple source elements let the browser choose a supported format.

```
<audio controls>
  <source src="song.mp3" type="audio/mpeg">
  <source src="song.ogg" type="audio/ogg">
</audio>
```

Attribute	Simple meaning
src	Path or URL of the media file.
type	MIME type such as audio/mpeg or video/mp4.

💡 **source = actual media file**

8. <video> Tag

The <video> element embeds video in a webpage.

```
<video controls poster="preview.jpg" width="640">
  <source src="movie.mp4" type="video/mp4">
  <source src="movie.webm" type="video/webm">
  Your browser does not support video.
</video>
```

Attribute	Simple meaning
controls	Shows video controls.
poster	Image shown before playback.
autoplay	Requests automatic playback; browsers may restrict it.
loop	Repeats the video.
muted	Starts without sound.
width	Displayed width.
height	Displayed height.
preload	Hints how much video data to load.

Output: a video player appears; the poster is shown before playback and the browser can choose a supported source.

💡 **poster = picture BEFORE video | source = media file TO PLAY**

Complete Multimedia Example

```
<h2>My Audio</h2>
<audio controls loop>
  <source src="music.mp3" type="audio/mpeg">
</audio>

<h2>My Video</h2>
<video controls poster="poster.jpg" width="640">
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
</video>
```

9. Quick Viva Revision

Question	Answer
Semantic tags?	Tags that describe the meaning/purpose of content.
header?	Introductory content.
nav?	Navigation links.
main?	Main unique content.
article?	Self-contained content.
section?	Related content.
footer?	Footer information.
required?	Compulsory value.
pattern?	Custom format check.
minlength vs min?	Characters vs value.
maxlength vs max?	Characters vs value.
type=email?	Email input.
type=tel?	Telephone input.

type=number?	Numeric input.
audio?	Embeds audio.
controls?	Shows media controls.
autoplay?	Requests automatic playback.
loop?	Repeats media.
video?	Embeds video.
poster?	Preview image before playback.
Multiple source tags?	Alternative media formats.

★ One-Page Memory Map

SEMANTIC: header | nav | main | article | section | footer

FORM: email | tel | date | textarea

VALIDATION: required | pattern | minlength | maxlength | min | max

AUDIO: audio | controls | autoplay | loop | source

VIDEO: video | controls | poster | source

💡 ~Yug Sardhara